

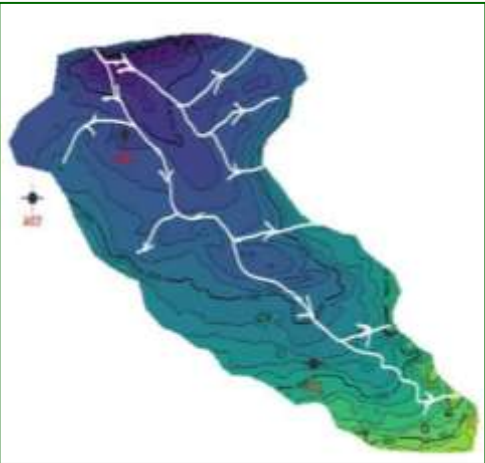


bp India Student Challenge 2025-26

Reservoir Revivers

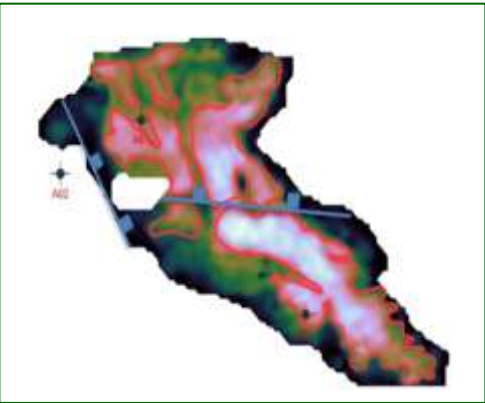
Where Insight Revives Energy

Dominant Channel Fairway



Thickness (Isopach) - NW is 100-135 m thick, SE is 20-30 m. Sediment physically cannot travel uphill to preferentially deposit at its source. Thick = proximal mouth (NW), thin = distal fringe (SE).

Valley Rule - Tributary branches join the main trunk *downslope*. All confluences on the white fairway map are consistent only with NW-high → SE-low geometry. Tributaries cannot climb to meet the trunk. this area the channel spreads out in



Combined Trap Type - Channel levees form the stratigraphic traps along the margins

The possibility of growth faults have been asserted on the bases of changes in elevation of turbidite SNA values, sharp turning of contours and well-log correlation

WELLS	SAND THICKNESS (Calculated from channel thickness)	WELL LOG THICKNESS	REASON
A01	99 m (Approx)	99.47 m	A1 sand thickness and well log thickness corroborate, because the contours are widely spaced and hence the dip in that region is almost horizontal thus the true sand thickness is same as the vertical thickness of the sand calculated via well log.
A02	Cannot calculate (Data Not Available)	95.1 m	A2 is outside the map area hence cannot be concluded
A03	70 m (Approx)	59.87 m	A3 sand thickness and well log thickness do not corroborate, because the contours are closely spaced and hence the dip in that region is high thus the true sand thickness is not same as the vertical thickness of the sand calculated via well log.

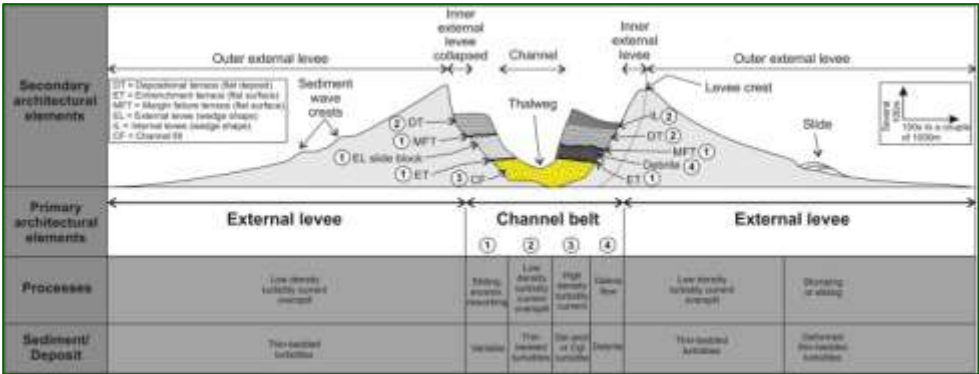
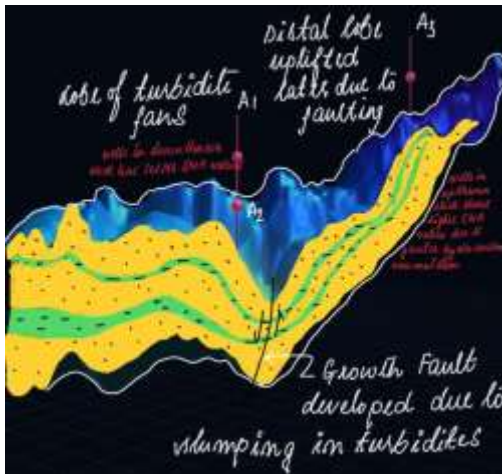
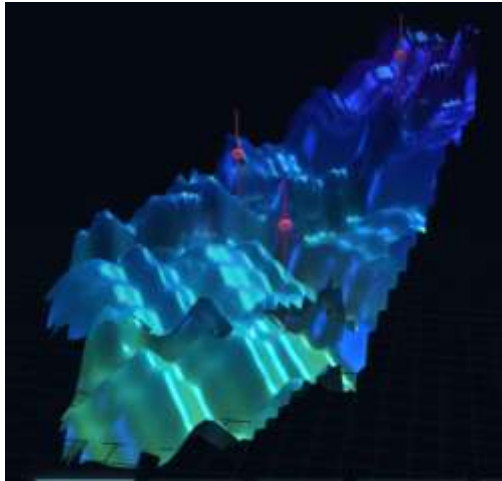


Figure: Conceptual model of deep-water channel-levee depositional architecture

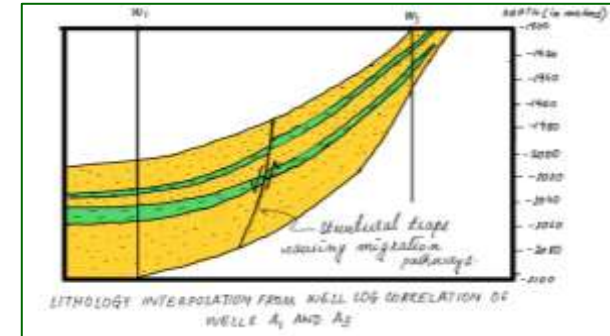
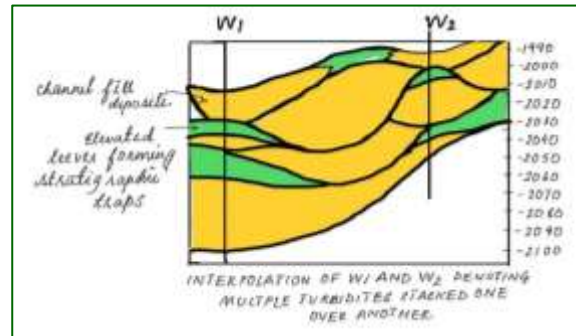
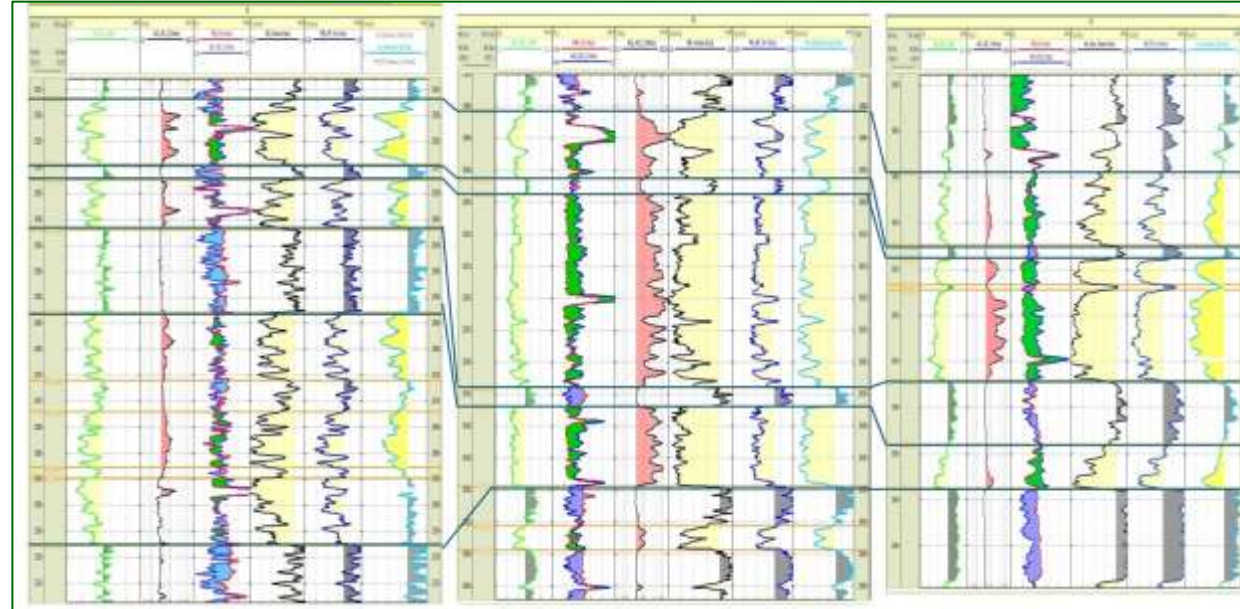
Modelling and Well Log Correlation



Well A01

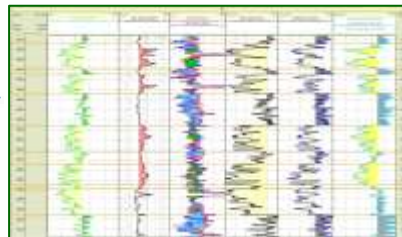
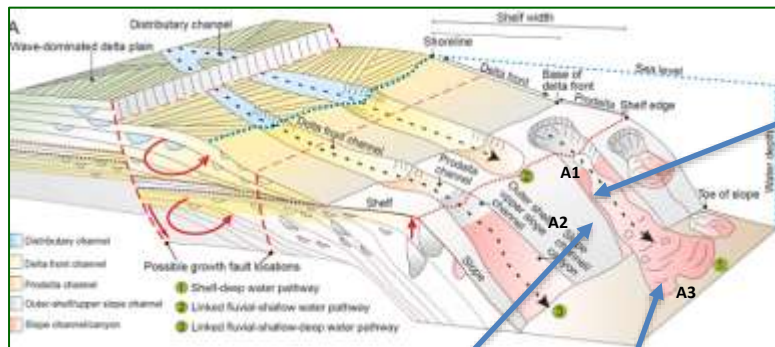
Well A02

Well A03



Reservoir Geomorphology and Depositional Environment

SHELF EDGE DEPOSITIONAL ENVIRONMENT DURING LOW STAND SYSTEM TRACK



Well A01 (Toe of Slope Turbidites): The log shows "serrated" or "thin-bedded" signatures, representing the **Bouma Sequence**. Each spike is a pulse of sand (turbidity current) followed by a return to baseline shale (quiet hemipelagic settling)



Well A02 (Slope Channel): The log shows high-amplitude, "blocky" sand signatures interrupted by shales. This is typical of **channel-fill deposits**, where high-energy flows leave thick sand packages within the canyon.



Well A03 (Distal Lobe Fringe (Off-Axis))
We hereby conclude that **A03** sits on the fringe of a laterally offset lobe. What arrives at **A03** is only the dilute, fine-grained flow tail — explaining the paradox of worse Bouma development despite being distally positioned.

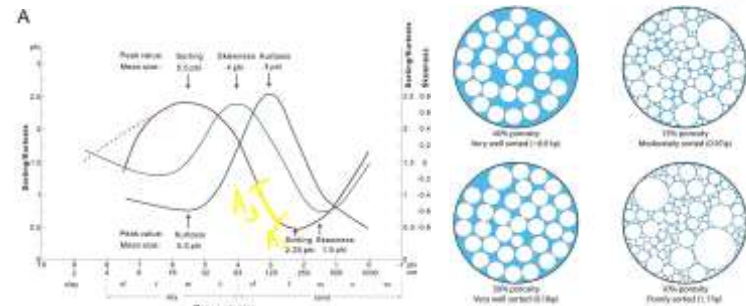
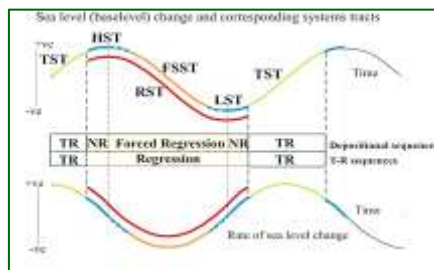


Image reference - Contourite porosity, grain size and reservoir characteristics by Xiaohang et al
Sedimentary Geology - Rocks, Environments and Stratigraphy by Michael Rygel and Page Quinton

- The reservoir geomorphology of a shelf-edge depositional setting is characterized by high-energy, progradation, and sometimes unstable systems that transfer massive amounts of sediment directly to the deep slope.
 - These settings, often active during low-stand or falling-stage sea levels, are dominated by **shelf-edge deltas** and associated gravity-flow deposits
 - Sea level fall with aggradation dominant shelf edge deltas and submarine lobe complex
- The shelf margin was continuously supplying sediment for the generation of turbidity currents in the shelf edge region

Well	Depositional Setting	ϕ -k Relationship	Formation process
A01	Toe-of-slope turbidites (Bouma sequences)	Significant clustering – similar porosity values produce nearly the same permeability due to very thin lamination and shale barriers	A1 is at the proximal part of turbidite so its location should have coarser grain size than that of A3. Hence better sorting and better clustering in porosity-permeability cross-plot than A3.
A02	Slope channel / channel-fill sands	Strong positive correlation between porosity and permeability due to well-connected pore networks	High-energy channel flow deposits tend to have thick sands as evinced from the well-log data and also from the porosity-permeability cross-plot.
A03	Distal lobe fringe (off-axis)	High porosity but low permeability due to clay-filled pores and small pore throats	At the distal lobe fringe part of turbidite, clay content starts to increase, also evident from the well-log data. Therefore, much less clustering in porosity-permeability cross-plot is seen.

Triple-Combo Log Display: Wells A1, A2, A3

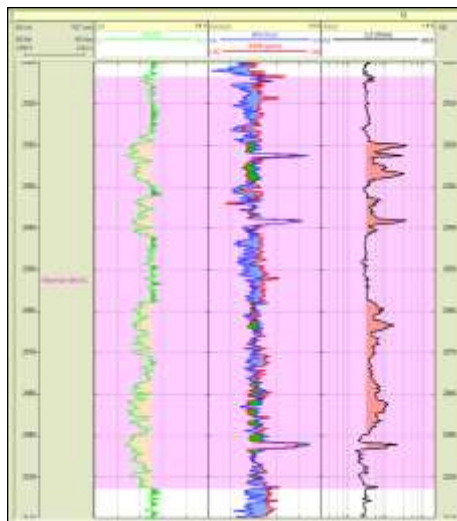


Fig. 1: Well Log Display for Well A01

Well A01 (2003.54–2103.01 m)

Low-moderate GR with spikes → sand-shale intercalations
N-D crossover → oil effect or lithology change
Elevated LLD → potential HC zones

Well A03 (1879.67–1939.54 m)

Higher GR than A1 and A2 → shalier reservoir
N-D crossover (~1918-1924 m) + high LLD → possible HC zone

Well A02 (1987.21–2082.31 m)

Low-Moderate GR → thick blocky sand
Strong N-D crossover (~2010-2020 m) → probable gas zone
High LLD throughout → excellent HC potential

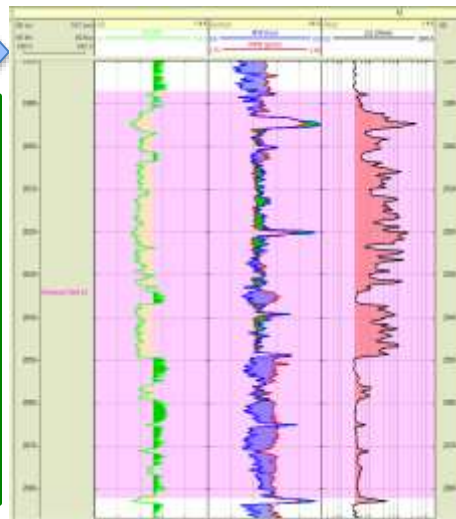


Fig. 2: Well Log Display for Well A02

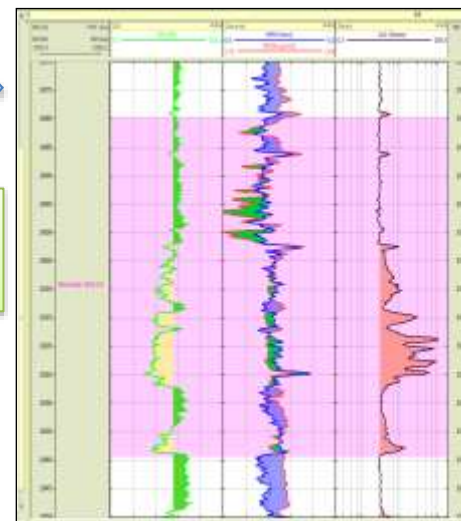


Fig. 3: Well Log Display for Well A03

Data Uncertainties:

- In well A03, there is a mismatch between the gamma ray log and the crossplot between the neutron and density log. There is a cross over in the region of high Gamma ray log value (high shale).
- In A02 (eg., at 1995m depth), we are getting near zero neutron and log in the region of low API value.
- Resistivity scale inconsistency between wells affects inter-well comparison
- N-D crossover interpretation limited without independent gas confirmation or fluid samples
- Lack of shallow resistivity prevents invasion assessment

Limitations:

- Near well A02, the data (maps) is very limited. Since A02, well log analysis gave favorable reservoir values (high NTG value).
- Single resistivity curve (LLD only) → invasion profile and true formation resistivity (R_t) cannot be validated
- No caliper log available → borehole condition unknown; density and neutron may be affected by washout
- Absence of PEF, sonic, NMR or mineralogy logs → lithology interpretation relies solely on GR trends
- Environmental corrections (temperature, calliper log mud weight, salinity) not verified → may affect calculated parameters.
- Due to intercalated sand and shale layer, it was a challenge to calculate the NTG because of sparse well data

Petrophysical Analysis

Volume of Shale

$$IGR = \frac{GR_{log} - GR_{min}}{GR_{max} - GR_{min}}$$

- GR_{log} : Measured gamma ray value at a given depth
- GR_{min} : Minimum gamma ray representing clean sand
- GR_{max} : Maximum gamma ray representing shale

Vshale was computed using IGR, Stieber, and Larionov methods; however, linear IGR was used for final evaluation because it provides a more conservative estimate where feldspathic/arkosic sands may show elevated GR, especially under the high Vsh cutoff of 80%.

Porosity

NPHI-porosity from neutron log

PHID-porosity from the density log

$$\phi_d = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

ϕ_d : Density porosity
 ρ_{ma} : Matrix density (g/cc)
 ρ_b : Bulk density from log
 ρ_f : Fluid density (g/cc)

A matrix density greater than 2.65 g/cc was used because the reservoir is not purely quartzose and likely contains feldspathic/arkosic grains and heavy minerals, which increase the effective matrix density.

$$\phi = \frac{\phi_n + \phi_d}{2}$$

ϕ : Combined porosity
 ϕ_n : Neutron porosity (from neutron log)
 ϕ_d : Density porosity (from density log)

$$\phi_e = \phi_t(1 - V_{sh})$$

ϕ_e : Effective porosity
 ϕ_t : Total porosity
 V_{sh} : Volume of shale

Water Saturation

Indonesian Model

$$SW_{Indonesia} = \left\{ \frac{\sqrt{\frac{1}{Rt}}}{\left(\frac{V_{sh}^{(1-0.5V_{sh})}}{\sqrt{Rsh}} \right) + \sqrt{\frac{\phi_e^m}{a \cdot Rw}}} \right\}^{(2/n)}$$

S_w : Saturation
 v_{sh} : Volume of Shale
 R_t : True Resistivity
 m : cementation factor
 ϕ_e : Effective Porosity
 R_w : Resistivity of Water
 a : Tortuosity factor
 R_{sh} : Resistivity of Shale

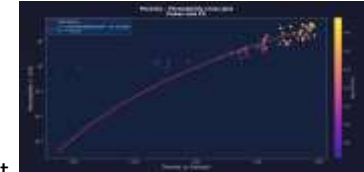
Indonesian model is used for calculating the water saturation because the study area has intercalated layer of shale and sand. This model gives reasonable range considering the shaly region

Permeability

Using the crossplots of porosity vs permeability

The permeability relationship was modeled using:

$$k = A \cdot \phi^B$$



Where:

- k = permeability (mD)
- ϕ = porosity (fraction)
- A = regression constant
- B = exponent controlling sensitivity of permeability to porosity

Constants A and B were determined using **log-log regression** of the core dataset.

Reservoir Deliverability Characteristics

Flow Capacity (F)

$$F = k \cdot h$$

Where:

k = permeability (mD)

h = net pay thickness (m)

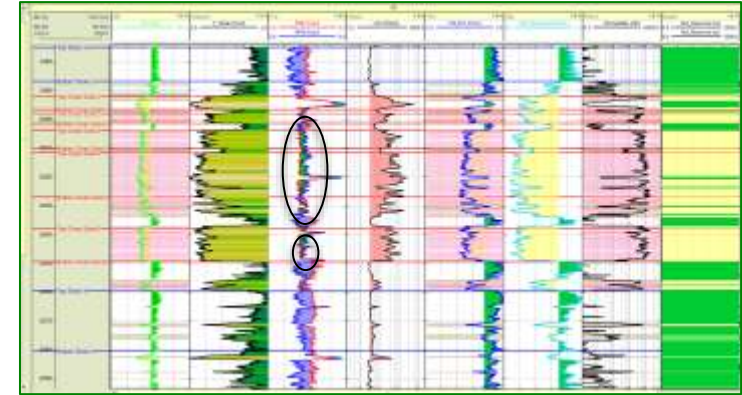
Reservoir Quality

Index (RQI)

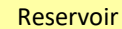
$$RQI = 0.0314 \sqrt{\frac{k}{\phi}}$$

RESERVOIR
REVIVERS

Well A02



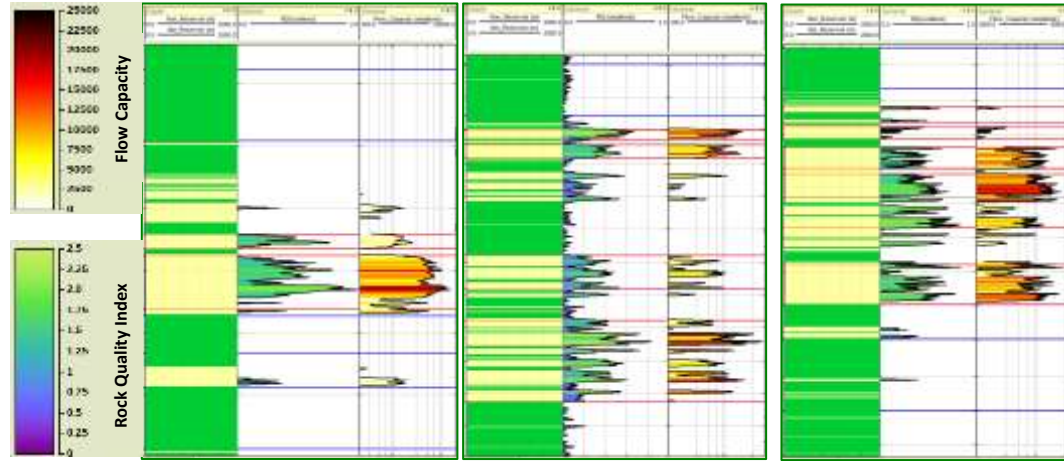
In similar manner, interpretations are done for all the wells and the values are added in the summary table



Reservoir Deliverability Characteristics and Petrophysical Properties Summary

Deliverability is fundamentally controlled by:

$$q \propto \frac{kh}{\mu} \Delta P$$



Parameter	Well A03	Well A01	Well A02
Reservoir Quality (RQI)	High, concentrated in main interval	Moderate to high in several intervals	Moderate, distributed
Flow Capacity (kh)	Strong peak in central sand	Multiple moderate peaks	Moderate peaks
Reservoir Architecture	Thick continuous sand	Stacked sands	Thin stacked sands
Vertical Heterogeneity	Moderate	Moderate-high	High
Deliverability	High	Moderate-High	Moderate

Petrophysical Properties Summary

Petrophysical Property	A01	A02	A03
Gross thickness (m)	99.47	95.10	59.87
Net Reservoir (m)	47.39	53.03	22.25
Net pay (m)	36.27	52.27	21.03
Net-to-Gross (%)	36.46	54.96	35.12
Avg Vsh	0.50	0.44	0.55
Avg Φ_{eff}	0.17	0.18	0.15
Avg Sw	0.64	0.45	0.61
Avg Permeability (mD)	77.34	330.09	123.95
Flow Capacity (kh)	455.02	5554.49	2267.58
Rock Quality Index (RQI)	0.35	0.75	0.35
Deliverability Ranking	3	1	2

Net-to-Gross Analysis

Gross Reservoir Thickness

For vertical wells, gross reservoir thickness is calculated as the difference between reservoir bottom and top depths.

Gross Thickness = Bottom depth - Top depth

Where,

- Bottom = Base of reservoir interval (m)
- Top = Top of reservoir interval (m)

Well A01: H1 = 2103.01 - 2003.54 = **99.47 m**

Well A02: H2 = 2082.31 - 1987.21 = **95.10 m**

Well A03: H3 = 1939.54 - 1879.67 = **59.87 m**

Net Reservoir Thickness

Net reservoir is calculated by applying petrophysical cutoffs to identify reservoir-quality intervals.

Petrophysical cutoffs: $V_{\text{shale}} \leq 80\%$, $\phi \geq 15\%$

Methodology:

1. Thickness log generated in RokDoc using $V_{\text{shale}} \leq 80\%$ and $\phi \geq 15\%$ cutoffs
2. Intervals meeting both criteria flagged as net reservoir
3. Exported to Excel and summed for each well's reservoir zone

Net Pay Thickness

Method: Same as net reservoir with additional $S_w \leq 70\%$ filtering applied.

Net to Gross Estimation

Ratio of net reservoir thickness to gross thickness, indicating the proportion of reservoir-quality rock

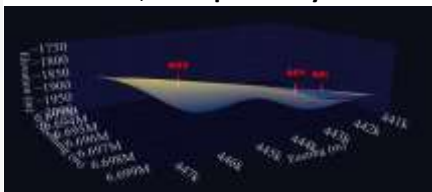
$$N/G (\%) = (\text{Net Pay thickness} / \text{Gross thickness}) \times 100$$

Reservoir Thickness Summary

Well	Gross Thickness (m)	Net Reservoir (m)	Net Pay (m)	N/G (%)
A01	99.47	47.39	36.27	36.46
A02	95.10	53.03	52.27	54.96
A03	59.87	22.25	21.03	35.12

Further development recommendation

- **Well A02** shows the most favorable **NTG value and petrophysical properties** from well log analysis.
- Geological studies and **well log correlations** indicate a **fault between Wells A02 and A01**.
- Therefore, an **exploratory well is recommended near A02**



However, **limited data in this region** restricts further detailed interpretation.

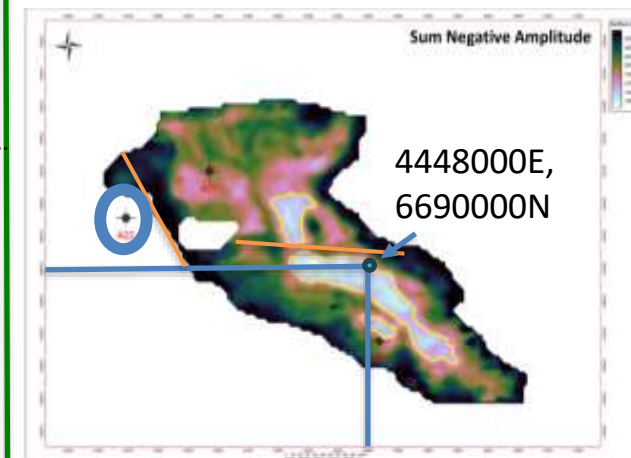
Net pay calculated using additional water saturation cutoff: $V_{\text{shale}} \leq 80\%$, $\phi \geq 15\%$, $S_w \leq 70\%$

Appraisal Well

The selected location is proposed for drilling the **next appraisal well** to evaluate the **size, quality, and commercial potential** of the hydrocarbon accumulation. The point lies within a **channel-levee depositional system** and in the **upthrown block**, providing a favorable trapping location.

Faults in the region may act as the **potential migration pathways**, allowing hydrocarbons to move from the **source rock to the reservoir** (away from proximal part). The **SNA attribute map** shows a **bright spot anomaly**, and the point lies near its boundary, helping to determine the **lateral extent of the reservoir**.

Therefore, the **appraisal well** is recommended near **444000E, 669000N**.



STOIIP Calculation I

STOIIP(MMBO):-

$$(\text{Bulk Rock Volume} * 7758 * \phi * (1 - s_w) * 1/B_0) * 1/10^6$$

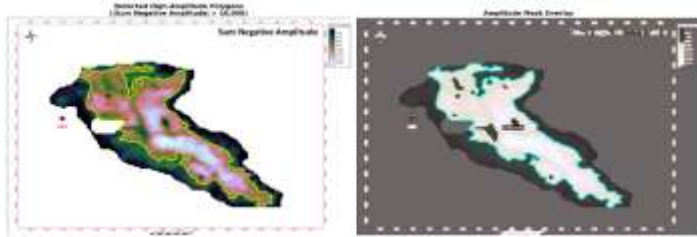
Bulk Rock Volume (acre feet) = A * h, A = Drainage area (acres), h = Net pay thickness (feet), 7,758 = API Bbl per acre-foot
= Porosity, Sw = water saturation, Bo = 1.3 Formation volume factor (Reservoir Bbl/STB) 1/Bo = Shrinkage (STB/reservoir Bbl)

We have calculated 3 STOIIP values in order to give a probabilistic estimation- min, most likely and max. Methods used for calculating the parameters are explained below

1.Area

1.1.Max:-

For this case we have taken the polygons which encloses the amplitude>6000

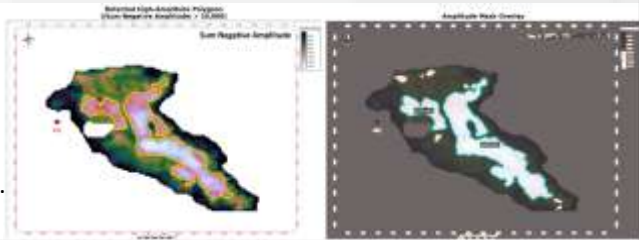


Area=7.983sq.Km

1.2.Most likely:-

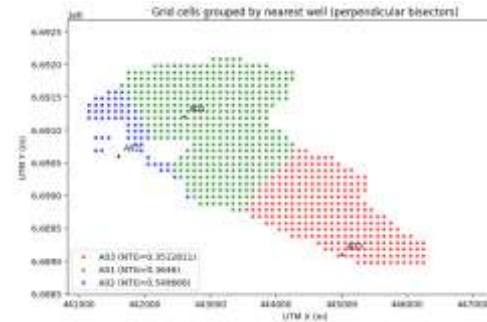
these polygons encloses the amplitude>10000.

Area=0.722+3.580 sq.
Km.= 4.302 sq.Km.



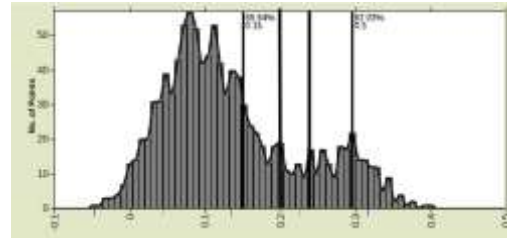
2. Average net pay thickness:-

Channel thickness values are digitized and extracted from the given channel thickness map by dividing it into 100X100m grids. We have used Voronoi partition by connecting the wells. We have 3 groups of channel thickness data, this is multiplied with the net to gross ratio of the respective wells. Mean of these net pay thickness is taken for h= 29ft (8.9m)



3. Petrophysical parameters:-

Porosity histogram plots
A01



For porosity and saturation values, we have taken 3 values for min, max and most likely values for each wells A01, A02 and A03 from their histograms.

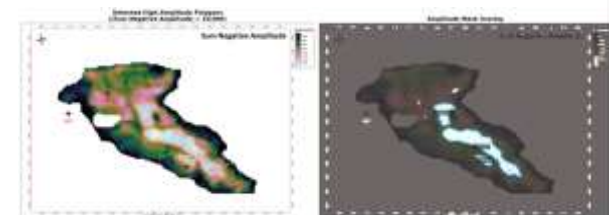
By using the Voronoi partition we have the count of the points belonging to each group and its respective wells.

Using this as the weights for the values we have calculated the weighted average for each models (min, max, most likely)

1.3.Min:-

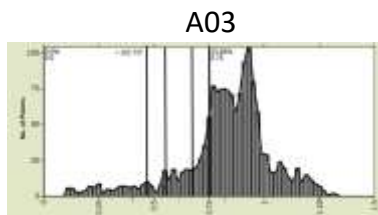
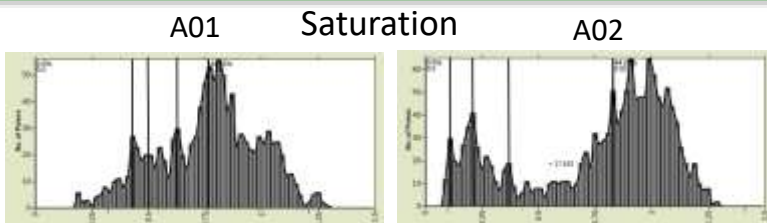
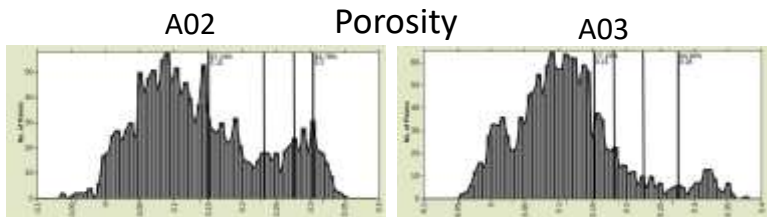
For this case we have taken the polygons which encloses the amplitude>16000

Area=0.306+1.123+0.098 sq.
Km=1.527 sq.Km



STOIP Calculation II

3. Petrophysical parameters:-



Weights for each well

UTM_X_m	UTM_Y_m	Thickness	Nearest_V	NTG	Net_Thickness_m	
441150	6691179	20	A02	0.549666	10.99332	A01 412
441150	6691279	10	A02	0.549666	5.49666	A02 60
441150	6691379	16.67	A02	0.549666	9.162932	A03 317
441250	6690679	0	A02	0.549666	0	
441250	6690879	0	A02	0.549666	0	

Porosity

Model	A01	A02	A03
Max	0.3	0.3	0.28
Most Likely	0.24	0.27	0.22
Min	0.2	0.23	0.18

Saturation

Model	A01	A02	A03
Max	0.45	0.11	0.47
Most Likely	0.5	0.21	0.55
Min	0.6	0.37	0.67

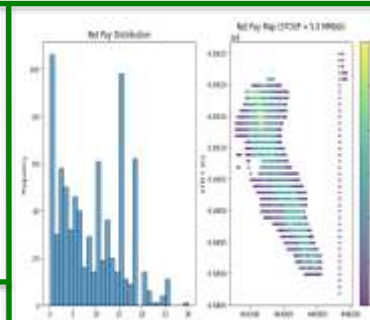
Weighted mean:-

$$(W1 \cdot P1 + W2 \cdot P2 + W3 \cdot P3) / (W1 + W2 + W3)$$

Where P-petrophysical properties

4. Final STOIP values:-

Model	ϕ	S_w	Area(km ²)
Max	0.2918	0.431	7.983
Most Likely	0.23417	0.49801	4.302
Min	0.19417	0.61077	1.527



Model	STOIP (MMBO)
Max	57.06
Most Likely	21.8
Min	4.97

STOIP CALCULATION	
Grid cells:	789
Total area:	377.3 acres (1.53 km ²)
Avg net pay:	8.9 m (29 ft)
Porosity ϕ :	0.19417
Water sat Sw:	0.61077
Oil FVF Boi:	1.3
STOIP = 4.97 MMbbl	